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The exterior derivative of multiplicative differential forms

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Summary

On a Lie groupoid, a multiplicative Ehresmann connection gives rise to a multiplicativity-preserving exterior derivative on differential forms, with values in a bundle of ideals. This turns out to be a particular case of the fact that the exterior derivative is a cochain map on the complex of differential forms on its nerve.

In the infinitesimal picture, we have a similar situation: given a so-called infinitesimal multiplicative connection on a Lie algebroid, it gives rise to an operator on infinitesimal multiplicative (IM) forms, also known as Spencer operators. We define these exterior derivatives below and present some interesting properties and applications.

Based on the work by Fernandes, Mărcuț (2023) and Crainic, Salazar, Struchiner (2015).

Preliminaries

- A *bundle of ideals* on a Lie groupoid $G \rightrightarrows M$ is a vector bundle $\mathfrak{k} \subset \ker\rho$, such that $\mathrm{Ad}\colon G \curvearrowright \ker\rho$ restricts to a representation of $\mathfrak{k}$. On the Lie algebroid A, this reads
$$[\alpha,\xi] \in \Gamma(\mathfrak{k}) \quad \text{for any } \alpha \in \Gamma(A) \text{ and } \xi \in \Gamma(\mathfrak{k}).$$
That is, $\mathfrak{k}$ is a subrepresentation of the adjoint "representations" of G and A.
- A *multiplicative (Ehresmann) connection* for a bundle of ideals $\mathfrak{k}$ is a distribution $E \subset TG$ which is multiplicative (i.e. a subgroupoid of $TG \rightrightarrows TM$) and satisfies $K \oplus E = TG$, where $K \subset TG$ is obtained from $\mathfrak{k}$ using left (or right) translations.
- Equivalently, a multiplicative Ehresmann connection E can be described using a differential form $\omega \in \Omega^1(G; s^*\mathfrak{k})$, satisfying:
  - ω is *multiplicative*: for any composable pair $(g,h) \in G^{(2)}$, there holds
$$(m^*\omega)_{(g,h)} = \mathrm{Ad}_{h^{-1}} \circ ({\mathrm{pr}_1}^*\omega)_{(g,h)} + ({\mathrm{pr}_2}^*\omega)_{(g,h)}. \tag{M}$$
  - ω restricts to the identity map on $\mathfrak{k}$, i.e. $\omega|_{\mathfrak{k}} = \mathrm{id}_{\mathfrak{k}}$.

  The correspondence is given by $E = \ker\omega$, and conversely, $\omega_g(v) = \mathrm{d}(L_{g^{-1}})_g(v^K)$.
- The notion of multiplicativity of a representation-valued form, as in (M) above, has an infinitesimal analogue: a $\mathfrak{k}$-valued *Spencer operator* on $A \Rightarrow M$ is a pair (L,l),
$$L\colon \Gamma(A) \to \Omega^k(M;\mathfrak{k}) \text{ is linear}, \qquad l\colon \Gamma(A) \to \Omega^{k-1}(M;\mathfrak{k}) \text{ is tensorial},$$
where the latter is called the *symbol* of L. They must satisfy the Leibniz rule
$$L(f\alpha) = fL(\alpha) + \mathrm{d}f \wedge l(\alpha), \tag{L}$$
and certain other compatibility conditions. Here k is called the *degree* of (L,l).
- Any multiplicative form ϑ can be linearized (differentiated) to a Spencer operator, in a similar fashion as representations. This is referred to as the *Lie functor*:
$$L(\alpha)_x(v_i)_i = \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \phi_\varepsilon^{\alpha^L}(1_x)\cdot\vartheta(\mathrm{d}(\phi_\varepsilon^{\alpha^L})_{1_x}(v_i))_i, \quad \text{with symbol} \quad l(\alpha) = u^*(\iota_\alpha\vartheta).$$
- An *infinitesimal multiplicative (IM) connection* is a Spencer operator $(\mathcal{C}, v)$ such that $v|_{\mathfrak{k}} = \mathrm{id}_{\mathfrak{k}}$. In other words, the symbol is just a splitting of the sequence
$$\begin{tikzcd} 0 \arrow[r] & \mathfrak{k} \arrow[r] & A \arrow[l, bend left, "v"] \arrow[r] & A/\mathfrak{k} \arrow[r] & 0. \end{tikzcd}$$
By the Leibniz identity (L), the principal part $\mathcal{C}$ hence splits into two parts: a linear connection ∇ on $\mathfrak{k}$ and a tensor field $U \in \Gamma(H^* \otimes T^*M \otimes \mathfrak{k})$, where $H = \ker v$,
$$\mathcal{C}(\alpha) = \nabla(v\alpha) + U(h\alpha).$$
A part of the information contained in U is that it measures the failure of $v\colon A \to \mathfrak{k}$ to be the splitting of Lie algebroids, not just of vector bundles.

Exterior derivative of multiplicative forms (groupoids)

Given a multiplicative connection ω on a Lie groupoid G, we straightforwardly define
$$\mathrm{D}^\omega\colon \Omega^\bullet(G;s^*\mathfrak{k}) \to \Omega^{\bullet+1}(G;s^*\mathfrak{k}), \quad (\mathrm{D}^\omega\vartheta)(v_i)_i = (\mathrm{d}^{\nabla^s}\vartheta)(h(v_i))_i.$$
Here, ∇ is the induced connection on $\mathfrak{k} \to M$ as above and $\nabla^s = s^*\nabla$. More directly,
$$\nabla_X\xi = v[h(Y), \xi^L]|_M,$$
for any $\xi \in \Gamma(\mathfrak{k})$, where $Y \in \mathfrak{X}(G)$ is any lift of $X \in \mathfrak{X}(M)$ along the source: $s_*Y = X$.

To obtain a conceptual viewpoint about why D^ω preserves multipicativity, note that multiplicative forms are precisely the forms in the kernel of the simplicial differential
$$\delta\colon \Omega^k(G;s^*\mathfrak{k}) \to \Omega^k(G^{(2)};(s\circ\mathrm{pr}_2)^*\mathfrak{k}), \quad \delta = \mathrm{pr}_2^* - m^* + \Phi_*\mathrm{pr}_1^* \tag{1}$$
where $\Phi\colon (s\circ\mathrm{pr}_1)^*\mathfrak{k} \to (s\circ\mathrm{pr}_2)^*\mathfrak{k}$ is the isomorphism $(g,h,\xi) \mapsto (g,h,\mathrm{Ad}_{h^{-1}}\xi)$. Hence, one rather proves that D^ω is a map of cochains (at degree one of the nerve):
$$\boxed{\delta\,\mathrm{D}^\omega = \mathrm{D}^\omega\delta} \tag{2}$$
where D^ω on the right-hand side is defined as $\mathrm{D}^\omega = h^*\,\mathrm{d}^{\nabla^{s\circ\mathrm{pr}_2}}$.

The exterior derivative as a cochain map

A similar identity to (2) holds at any degree of the nerve. Diagramatically:
$$\begin{tikzcd} \Omega^{\bullet+1}(M;\mathfrak{k}) \arrow[r, "\delta^0"] & \Omega^{\bullet+1}(G;s^*\mathfrak{k}) \arrow[r, "\delta^1"] & \Omega^{\bullet+1}(G^{(2)};(s\circ\mathrm{pr}_2)^*\mathfrak{k}) \arrow[r, "\delta^2"] & \cdots \\ \Omega^{\bullet}(M;\mathfrak{k}) \arrow[r, "\delta^0"] \arrow[u, "\mathrm{d}^\nabla"] & \Omega^{\bullet}(G;s^*\mathfrak{k}) \arrow[r, "\delta^1"] \arrow[u, "\mathrm{D}^\omega"] & \Omega^{\bullet}(G^{(2)};(s\circ\mathrm{pr}_2)^*\mathfrak{k}) \arrow[r, "\delta^2"] \arrow[u, "\mathrm{D}^\omega"] & \cdots \end{tikzcd}$$
Here, we are denoting by δ^ℓ the simplicial differential at an arbitrary degree ℓ of the nerve, defined as the alternating sum over the $\ell+1$ face maps, similarly as in equation (1), and $\mathrm{D}^\omega = h^*\,\mathrm{d}^{\nabla^{s\circ\mathrm{pr}_\ell}}$. As a consequence, D^ω descends to a map in $\delta^\bullet$-cohomology:
$$\mathrm{D}^\omega\colon H^{\ell,k}(G;\mathfrak{k}) \to H^{\ell,k+1}_{\mathrm{Hor}}(G;\mathfrak{k}).$$
The subsript Hor stands for the *horizontal subcomplex* of the complex above, consisting of forms on the nerve which vanish whenever a vector from $K^{(\ell)} = K^\ell \cap TG^{(\ell)}$ is inserted.

Exterior derivative of Spencer operators (algebroids)

We now enter the infinitesimal realm. Defining the exterior deriative of a Spencer operator (L,l) is more convoluted: given an IM connection $(\mathcal{C},v)$ on $A \Rightarrow M$, the exterior derivative of (L,l) is the Spencer operator $\mathrm{D}^{(\mathcal{C},v)}(L,l)$, given by
$$\mathrm{D}^{(\mathcal{C},v)}(L,l)(\alpha) = (\mathrm{d}^\nabla L\alpha - \underline{L}\,\dot\wedge\,\mathcal{C}\alpha, \underline{L}h\alpha),$$
where we have denoted $\underline{L} := L - \mathrm{d}^\nabla l$ which is a bundle map $A \to \Lambda^k(T^*M)\otimes\mathfrak{k}$ by the Leibniz identity for (L,l). We have also introduced $\underline{L}\,\dot\wedge\,\mathcal{C}\alpha \in \Omega^{k+1}(M;\mathfrak{k})$ as
$$(\underline{L}\,\dot\wedge\,\mathcal{C}\alpha)(X_0,\dots,X_k) = \textstyle\sum_i (-1)^i \underline{L}(\mathcal{C}\alpha(X_i))(X_0,\dots,\hat{X}_i,\dots,X_k).$$

This notion of the exterior derivative for Spencer operators is indeed the "right" one: if A is integrable to G and $(\mathcal{C},v)$ comes from a connection ω on G, we get a square:
$$\begin{tikzcd} \Omega^\bullet_m(G;s^*\mathfrak{k}) \arrow[r, "\mathrm{D}^\omega"] \arrow[d, "\mathrm{Lie}"'] & \Omega^{\bullet+1}_m(G;s^*\mathfrak{k}) \arrow[d, "\mathrm{Lie}"] \\ \Omega^\bullet_{im}(A;\mathfrak{k}) \arrow[r, "\mathrm{D}^{(\mathcal{C},v)}"'] & \Omega^{\bullet+1}_{im}(A;\mathfrak{k}) \end{tikzcd}$$
An analogous infinitesimal notion to the forms on higher degrees of the nerve is yet to be established. However, we still get a similar identity to the square at degree zero:
$$\delta^0\,\mathrm{d}^\nabla = \mathrm{D}^{(\mathcal{C},v)}\delta^0,$$
where $\delta^0\colon \Omega^\bullet(M;\mathfrak{k}) \to \Omega^\bullet_{im}(A;\mathfrak{k})$ is given by $(\delta^0\gamma)(\alpha) = (\mathscr{L}^A_\alpha\gamma, \iota_{\rho\alpha}\gamma)$ for all $\alpha \in \Gamma(A)$.

Curvature

- On groupoids, the *curvature* of a multiplicative Ehresmann connection ω is the multiplicative 2-form $\Omega^\omega = \mathrm{D}^\omega\omega$. It satisfies the *Bianchi identity* $\mathrm{D}^\omega\Omega^\omega = 0$.
- The *curvature* of an IM connection $(\mathcal{C},v)$ is the IM 2-form $\Omega^{(\mathcal{C},v)} = \mathrm{D}^{(\mathcal{C},v)}(\mathcal{C},v)$, i.e.
$$\Omega^{(\mathcal{C},v)}\alpha = \big(R^\nabla\cdot v\alpha + \mathrm{d}^\nabla(Uh\alpha), Uh\alpha\big).$$
Analogously to groupoids, the *infinitesimal Bianchi identity* holds: $\mathrm{D}^{(\mathcal{C},v)}\Omega^{(\mathcal{C},v)} = 0$.

Applications: affine deformations & curvings

- The space of IM connections for $\mathfrak{k}$ is an affine space over *horizontal* IM 1-forms (L,l), i.e. $l|_{\mathfrak{k}} = 0$. A natural question is thus how the curvature changes with an affine deformation $(\mathcal{C},v) \to (\mathcal{C},v) + \lambda(L,l)$. This is answered by the following equation:
$$\Omega^{(\mathcal{C},v)+\lambda(L,l)} = \Omega^{(\mathcal{C},v)} + \lambda\,\mathrm{D}^{(\mathcal{C},v)}(L,l) + \lambda^2\mathbf{c}_2(L,l),$$
where $\mathbf{c}_2$ maps to horizontal IM 2-forms: $\mathbf{c}_2(L,l)(\alpha) = -(L|_{\mathfrak{k}}\,\dot\wedge\,L\alpha, L|_{\mathfrak{k}}\cdot l\alpha)$.
- An IM connection is *curved* if its curvature is cohomologically trivial: $\Omega^{(\mathcal{C},v)} \in \mathrm{im}\,\delta^0$. A 2-form $F^{(\mathcal{C},v)} \in \Omega^2(M;\mathfrak{k})$ that satisfies $\Omega^{(\mathcal{C},v)} = \delta^0F^{(\mathcal{C},v)}$ is called a *curving* of $(\mathcal{C},v)$. The *curvature 3-form* of a curving is then defined as $G^{(\mathcal{C},v)} = \mathrm{d}^\nabla F^{(\mathcal{C},v)}$.
- A curving is determined up to an invariant 2-form $\gamma \in \ker\delta^0 \subset \Gamma(\Lambda^\bullet(T\mathcal{F})^\circ; z(\mathfrak{k}))$. Moreover, the following holds:
  - The curvature tensor of ∇ relates to the curving by $R^\nabla\cdot\xi = [\xi, F^{(\mathcal{C},v)}]$, $\forall\xi \in \Gamma(\mathfrak{k})$.
  - The Bianchi identity translates to the following identities for the 3-curvature:
$$\delta^0G^{(\mathcal{C},v)} = 0 \quad \text{and} \quad \mathrm{d}^\nabla G^{(\mathcal{C},v)} = 0.$$
  Hence $G^{(\mathcal{C},v)}$ is center-valued, transversal, and vanishes wherever $\operatorname{codim}\mathcal{F} \leq 2$.
  - The simplicial differential δ^0 relates to $\mathbf{c}_2$ as $\mathbf{c}_2(\delta^0\gamma) = -\frac{1}{2}\delta^0[\gamma,\gamma]$. Hence, the set of curved connections is an affine space over coh. trivial 1-forms $\mathrm{im}\,\delta^0$, and we have
$$F^{(\mathcal{C},v)+\delta^0\gamma} = F^{(\mathcal{C},v)} + \mathrm{d}^\nabla\gamma - \frac{1}{2}[\gamma,\gamma] \quad \text{and} \quad G^{(\mathcal{C},v)+\delta^0\gamma} = G^{(\mathcal{C},v)}.$$
  - If $\mathfrak{k}$ has a semisimple typical fibre, all IM connections are uniquely curved.